

Muhammad Fahad Waqar

Data Science — Data Engineer — Electrical Engineer

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Summary

As a Data Science and AI graduate student with a background in Electrical and Computer Engineering, I bring a strong foundation in data analysis, predictive modeling, and workflow automation. My experience spans developing data pipelines, designing analytical dashboards, and using tools like Python and SQL to extract actionable insights. Dedicated to solving real-world challenges through innovative, data-driven approaches, I aim to make a meaningful impact at the intersection of AI, data engineering, and sustainability.

Skills

- **Programming Languages:** Python, C/C++, SQL, Flutter, HTML, CSS
- **Libraries:** Pandas, Numpy, TensorFlow, OpenCV, Matplotlib, Keras
- **Tools:** Git, Bash, Android Studio, Jupyter Notebooks, Vivado, Power BI, Tableau, Metabase, MATLAB, Proteus Suite, Office Suite, Adobe Suite

Experience

Data Analyst, AI

Motive

Islamabad, Pakistan

March 2025 - August 2025

- Explore large datasets with SQL and Python to uncover AI trends and performance issues.
- Analyze AI model metrics (accuracy, precision, F1 score) to identify improvement areas.
- Design and run A/B tests and experiments to validate AI model changes and support feature development.
- Collaborate with QA, Product, and Data teams to support AI improvements and issue resolution.

Project Support Officer

Andromeda Technologies

Islamabad, Pakistan

December 2024 - March 2025

- Tracked milestones and monitored progress for the ATV Network Locking project from a PMO perspective.
- Created and maintained project dashboards and Gantt charts to visualize timelines and dependencies.
- Coordinated with internal teams and vendors through follow-ups to ensure timely task completion.
- Supported end-to-end delivery from requirement confirmation to Google/Netflix certification.

Data Engineer

SkyElectric

Islamabad, Pakistan

June 2024 - December 2024

- Extracted insights using advanced Python data analysis, improving decision-making.
- Designed and maintained systems dashboards and data visualizations using Metabase and Power BI.
- Managed and optimized PostgreSQL/MySQL databases for efficient storage, retrieval, and performance.
- Streamlined ETL processes, automated Linux workflows and managed CI/CD pipelines with Jenkins.

Intern, Digital Technologies

Zong CMPak

Islamabad, Pakistan

July 2023 - August 2023

- Leveraged SQL for efficient database management, data extraction, and query optimization.
- Applied data analysis techniques to uncover insights and extract meaningful information from datasets.
- Gained hands-on experience in applying machine learning models to solve real-world problems.
- Designed data visualizations with Power BI and Tableau, and performed sentiment analysis on textual data.

Projects

Real-Time Log Anomaly Detection Pipeline (AWS MLOps)

- Built a cloud-native AWS MLOps pipeline for real-time log anomaly detection using streaming ingestion, automated feature engineering, unsupervised ML models, and rolling retraining with drift monitoring.

Weekly Options Screener with ML Price Prediction

- Developed a full-stack financial decision support platform for weekly options income strategies integrating live screening, ML-based price prediction, backtesting, and portfolio-aware recommendations.

Claim Contradiction Detection over Knowledge Graphs

- Designed a multi-stage NLP pipeline combining knowledge graphs, semantic retrieval, NLI filtering, and LLM reasoning to detect intra-document contradictions at scale.

Hourly Stock Direction Prediction (NVIDIA)

- Developed an hourly stock direction prediction system using multi-source financial data and tree-based machine learning models, validated through feature ablation and deployment-aware model selection.

Machine Learning-Based Phishing Website Detection System

- Designed and deployed a production-ready phishing detection system using supervised machine learning on multi-source URL, DNS, and HTML features, achieving near state-of-the-art performance with optimized LightGBM models and a secure FastAPI-SvelteKit architecture.

AI-Generated Image Detection: ML vs Deep Learning Comparison

- Investigated the effectiveness of traditional machine learning models versus convolutional neural networks for detecting AI-generated images through systematic feature extraction, model training, and comparative performance evaluation.

Culturally Adaptive AI Advisor for Counseling Support (HCI Study)

- Designed and evaluated an HCI framework using AI-simulated patients and culturally adapted LLM-based suggestions to assess and improve counselors' cross-cultural competence through controlled user studies.

OBD Sentinel: IoT-based Automobile OBD Monitoring System

- Developed an IoT-based OBD Monitoring System with real-time vehicle diagnostics and cloud-based analytics. Designed hardware and a mobile app to deliver insights on performance, safety alerts, and maintenance trends.

Education

Asian Institute of Technology (AIT), Khlong Luang, Pathum Thani, Thailand

August 2025 – May 2027

MS Data Science & Artificial Intelligence

National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad, Pakistan

August 2020 – June 2024

BS Electrical Engineering – Major: Computer Engineering

- **Final Year Project:** IoT-based Automobile OBD Monitoring System

Achievements

- **Huawei ICT Competition:** 3 time National Finalist (2021, 2022, 2023).
- **Dean's Honor List at FAST-NUCES (Fall 2023)**
- **Fast Engineering Society:** Head Team Technical Writing (2023), Coordinator Team Seminars (2022)
- **IEEE FAST Student Chapter:** Coordinator Team Technical Writing (2023)
- **ISYWSC 22':** Member Team Media (2022)
- **Usher:** Openhouse 22', FAST-NUCES Islamabad